

**Vivo Y51S**

Vivo has launched a new midrange phone in China with the Exynos 880 5G processor and 4500 mAh battery. <https://digit.in/aug20-55>

**Huawei FreeBuds 3i**

Watch out for these new wireless earbuds from Huawei, which will be on sale on Amazon soon. <https://digit.in/aug20-56>

HP Omen 15 2020

Damn near perfect

HP experimented quite a lot with its Omen series of laptops and this year is no different. While the usual slew of generational upgrades continue to be present, the new Omen 15 is actually quite a different device in comparison to what has come before it. Let's see how it fared in our review.

Build and Design

The new HP Omen 15 super sleek and elegant. Almost every aspect of the Omen 15 is rock solid, except one; the Lid. The lid has far too much flex, it wobbles and the hinges don't retain the straight line alignment.

When it comes to the ports, on the right, you will find two USB 3.0 ports, separated from the DisplayPort and Thunderbolt 3.0 port by an exhaust vent. On the left, we get a full-sized SD Card reader, a combined headphone/microphone jack, a full size HDMI port and another USB 3.0 port.

Performance

The Omen we got houses a 1TB Samsung MZVLB1TOHBLR OEM drive which uses Samsung's Phoenix controller. The specification page lists the drive as having 3500MB/s and 3000MB/s sequential read-write speeds respectively. Using CrystalDisk Mark 7, we were able to verify that the drive indeed does almost reach those theoretical speeds.

The HP Omen 15 2020 has a 1080p IPS panel with a refresh rate of 144Hz. The panel supports 8-bit colour per channel, allowing for accurate and wholesome sRGB coverage. The panel is also pretty bright, registering 310 lux of brightness in the center, however, the corners were a little bit dimmer. The matte coating on the display handles reflections really well.

We fired up our standard suite of games for benchmarking the HP Omen 15's hardware and there were definitely



some very interesting findings. Most games stay well below the 100 fps mark, not really being able to take full advantage of the 144Hz panel. However, most games do run well above the 60 fps mark, meaning having that fast refresh rate panel is definitely going to improve your gaming experience.

For the creators out there, we've got render times from both Adobe Lightroom Classic and Premiere Pro. We are using 500 RAW files from a Nikon D850 and exporting them into JPG files in batches of 50, 100 and 500. During this time, we note the time taken for the machine to complete the task along with recording the thermals and CPU core states to see if there is any throttling happening. For video creators, we export a standard 4K timeline loaded with plenty of effects, transitions and LUTs into 4K H.264 files. Our export files have a duration of 5 minutes and 20 minutes.

During the 20 minute 4K video export, we note that the CPU never hits its advertised 5GHz boost on single core, or the 4.8GHz Intel Turbo Boost Max Frequency. While there was still thermal headroom during these operations, the power draw seems to suggest that the CPU was hitting the 45W TDP, especially during gaming. Premiere

tends to shift render load between the CPU and GPU rather aggressively, preventing a strain on either of those components. Regardless, the Omen 15 completes the render tasks assigned to it at a surprisingly fast pace.

Our testing was conducted in 24-28 degree ambient temperature and not once did the HP Omen 15 feel like it could double up as a toaster. The WASD keys recorded a temperature of 38-degrees, while the center of the keyboard was at 44-degrees. All readings were taken during our game-testing. The base is still too hot to put in your lap, but the fact that the rest of the machine doesn't have you reaching for an ice pack is definitely a delight.

There are two design choices that help the Omen 15 keep its cool. The first is the large perforations on the bottom lid that help the two fans suck in air with as little obstruction as possible. The second is the fin stack that runs all the way from the left to the right.

**Bottomline**

The HP Omen 15 2020 is a rock-solid contender for the next gaming laptop you buy. It packs the perfect hardware configuration, only thrown off by the choice of the Nvidia GeForce GTX 1650Ti. The display is bright and very colour accurate. Perhaps the most impressive are the thermals, which HP has done a really good job of keeping in check. There's also a user-upgradeable RAM option, with an additional, empty NVMe slot to add more high-speed storage. The only flaw is the lid, which wobbles and flexes. If you can live with a poor quality lid, then there's really no reason for you to not consider the HP Omen 15 as your next gaming laptop.

—Swapnil Mathur

SPECIFICATIONS

PROCESSOR: Intel Core i7-10750H (6 cores, 12 threads) | RAM: 16GB DDR4 @ 3200MHz | STORAGE: Samsung 1TB NVMe | GPU: Nvidia GeForce GTX 1650Ti | DISPLAY: Full HD IPS display with 144Hz refresh rate

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